115-2 新開課程規劃（BFN 學程）

115-2 新開課程整體規劃（BFN 學程，全英語授課）

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Plant Biology 植物生物學

- Course description
- Learning outcomes
- Prerequisites
- Textbooks and readings
- Weekly schedule (16 weeks)
- Assessment
- Course policies
- 設計理由與備課備忘（中文，自用）

Organelle Biology 胞器生物學

- Course description
- Learning outcomes
- Prerequisites
- Textbooks and readings
- Weekly schedule (16 weeks)
- Assessment
- Course policies
- 設計理由與備課備忘（中文，自用）

Plant Lipids 植物脂質

- Course description
- Learning outcomes
- Prerequisites
- Textbooks and readings
- Weekly schedule (16 weeks)
- Assessment
- Course policies
- 設計理由與備課備忘（中文，自用）

Plant Lipids: Reading list built on the Nakamura lab publications

- Wk 1 · Introduction: lipid diversity and why it matters

Wk 3 · Glycerolipid assembly, phosphatidic acid phosphatases, and ER–plastid cooperation

Wk 4 · Chloroplast lipids and membrane remodeling under phosphate starvation

Wk 5 · Phospholipid headgroups: phosphatidylcholine and phosphatidylethanolamine biosynthesis

Wk 6 · Desaturation, membrane fluidity, and the ER

Wk 9 · Storage lipids: triacylglycerol, lipid droplets, and algal oil

Wk 12 · Lipid signaling I: phospholipases, phosphatidic acid, and phosphoinositides

Wk 13 · Lipid signaling II: lipids in flowering, development, and ER homeostasis

Wk 14 · Lipid analysis and data-reading workshop

Wk 15 · Plant oils for food, nutrition, and industry

Candidate papers for the final presentation (one per pair; first come, first served on NTU COOL)

115-2 新開課程整體規劃（BFN 學程，全英語授課）

> 撰寫日期：2026-10-24。本檔為備課用的中文規劃；三門課的英文大綱分別在同目錄的三個檔案。
> 所有標示「待確認」的項目，請以臺大教務處 115 學年度行事曆與學程辦公室公告為準。

1. 三門課一覽

課程	年級	時段	節數／學分	大綱檔案
Plant Biology 植物生物學	大一	週一 6–8 節（13:20–16:20）	3 節／3 學分（假設）	`plant-biology.md`
Organelle Biology 胞器生物學	大二	週五 6–7 節（13:20–15:10）	2 節／2 學分（假設）	`organelle-biology.md`
Plant Lipids 植物脂質	大三	週五 8–9 節（15:30–17:20）	2 節／2 學分（假設）	`plant-lipids.md`；文獻清單 `plant-lipids-nakamura-reading-list.md`

待確認：課號、學分數、必修／選修、修課人數上限、教室（週一的課若要做顯微鏡觀察，需有可移動桌椅或實驗桌）。

本機閱讀版：`115-2-course-plan.html`（瀏覽器直接開）與 `115-2-course-plan.pdf`，由上述 Markdown 以 pandoc 產生；改了 Markdown 後需重新產生。

116-1 新課草案：`../116-1/plants-on-your-plate.md`（大一上，1 學分）、`../116-1/cellular-dynamics.md`（大二上，2 學分）。

2. 縱向設計：一條從細胞到分子的階梯

三門課加上 115-1 已開的兩門，構成 BFN 學程裡一條完整的植物／細胞生物學路徑：

```
大一下   Plant Biology        整株植物：細胞、組織、器官、生理、生殖、多樣性、應用
   ↓
大二下   Organelle Biology    拆開細胞：各胞器的結構、生合成、蛋白質運送、胞器溝通
   ↓
大三下   Plant Lipids         深入分子：脂質合成、膜、儲存、訊號、分析方法、油脂生技
   ↓
```

```
大三／四上  Introduction of Protein Homeostasis（115-1, BFN5013）
研究所      Plant Physiology and Stress Responses（115-1, BIOT7035）
```

設計原則：

- 各年級一個尺度：大一談整株植物與器官，大二談胞器，大三談分子。同一主題（例如葉綠體、內質網）在三門課都出現，但深度逐級加深，避免重複講同一張投影片。
- 每門課都連回學程主軸：BFN 是「生物科技 + 食品營養」，每門課每週至少一個食品／生技的例子（例如大一的「盤子裡的植物」、大二的粒線體與營養代謝、大三的食用油脂肪酸組成）。
- 每門課都連回你的研究：ER 恆定（大二第 5 週）、脂質調控 ER 壓力（大三第 13 週）、醣基化與蛋白質品管（大二第 5–6 週）。這些週的投影片可直接從實驗室 seminar 與 115-1 教材改寫。
- 第一輪開課的特殊狀況：115-2 的大二、大三學生沒有修過前一階的課（例如大三修植物脂質的人沒上過胞器生物學）。第一年每門課的前 1–2 週要補基礎；從 116 學年起才會是完整的階梯。

3. 學期行事曆與假日（待確認）

- 開學：2027-02-22（週一）上課開始，共 16 週，第 16 週為 6 月 7–11 日。來源：臺大教務處 115 學年度行事曆（行政會議通過版，教務處頁面另有 2026-09-08 更正版，我無法直接開啟臺大網域，請自行核對）。
- 可能停課的上課日：

日期	影響課程	原因	狀態
3/1（一）	Plant Biology 第 2 週	228 和平紀念日（2/28 週日）補假	待確認
4/5（一）	Plant Biology 第 7 週	清明節	幾乎確定
4/6（二）	無	兒童節（4/4 週日）補假	不影響
4/30（五）	Organelle Biology、Plant Lipids 第 10 週	勞動節（5/1 週六）補假	待確認：請查行事曆是否列為全校放假
6/9（三）	無	端午節	不影響

結論：週一的課只有 **14** 次上課，週五的課 **15** 次。三份大綱都已依此排進度；若 3/1 或 4/30 照常上課，該週預設為複習／彈性週，不需要重排。

4. 共同的評量與課程政策

三門課用同一套骨架，學生在不同年級看到一致的規則，你也只需維護一份政策文字：

- 評量結構：期中 30% + 期末（筆試或報告）30% + 平時作業 20–25% + 參與 10–15%。大一用兩次筆試，大三用文獻報告取代期末筆試，大二居中。
- 平時作業：大一是每週 studio 學習單與小考；大二是 4 次問題集；大三是閱讀回應與問題集。
- 出席與參與：點名不佔分，參與分數看課堂活動與討論。
- 生成式 **AI**：允許用於預習、整理筆記、修改英文；作業須註明使用方式；考試不得使用。這條請你決定是否採用。
- 平台：NTU COOL 放投影片、作業、公告；每週上課前一天上傳投影片。
- 期中回饋：第 8 週做 5 分鐘匿名問卷，第 9 週回應。

5. 備課時程建議（**2026-10 → 2027-02**）

時間	工作	備註
10 月底	向學程辦公室確認課號、學分、必選修、人數、教室、新開課程送審截止日	新開課程通常需學程課程委員會通過，請確認 115-2 的送審時程
11 月	定稿三份英文大綱，填入臺大課程網（課程大綱）	初選通常在 1 月上旬，大綱要在那之前上線
11–12 月	Plant Lipids 投影片（你最熟，先做完可以建立節奏）	每週 2 節，約 15 份投影片
12 月–1 月	Organelle Biology 投影片	第 5 週（ER）、第 13 週（葉綠體）可改寫 115-1 教材
1–2 月	Plant Biology 前 8 週投影片與 studio 活動材料	大一人數可能最多，活動材料要先試做
2/22 開學後	保持領先兩週，每週固定一個半天備下週	週五連上 4 節，建議週四晚上不要排其他事

可重用的 115-1 教材（粗估可省三成的製作時間）：

- Plant Physiology and Stress Responses（BIOT7035）的水分、礦物營養、光合、荷爾蒙、逆境各週 → Plant Biology 第 5、6、8、10、14 週，降低深度後使用。
- Introduction of Protein Homeostasis（BFN5013）的 ER／分泌途徑、UPR、自噬各週 → Organelle Biology 第 5、9 週。
- 實驗室 seminar 與你的 JXB 2022 lipid viewpoint 文章 → Plant Lipids 第 13 週。

Plant Biology 植物生物學

115-2 · Course code TBD · 3 credits (assumed) · Required/Elective TBD · Mon 6–8 (13:20–16:20) · Taught in English

Bachelor Program of Biotechnology and Food Nutrition (BFN) · Year 1 · Instructor: Yueh Cho

Course description

Plants feed us, clothe us, and run the planet's carbon cycle. This course gives first-year students a working map of how plants are built and how they work: from the plant cell and its unique organelles, through tissues and organs, to water and nutrient transport, photosynthesis, hormones, environmental sensing, reproduction, and evolutionary diversity. Every week links a core concept to a plant on your plate or a tool in the biotechnology industry. The course prepares you for Organelle Biology (year 2), Plant Lipids (year 3), and the plant-related electives of the BFN program.

Each week combines two lecture periods with one "studio" period for hands-on observation, problem solving, and short quizzes.

Learning outcomes

By the end of the course, you will be able to:

1. Describe the structure of the plant cell, the major tissue types, and the anatomy of roots, stems, leaves, flowers, seeds, and fruits.
2. Explain how plants move water, minerals, and sugars, and how they fix carbon in C3, C4, and CAM photosynthesis.
3. Outline how hormones and environmental signals (light, gravity, day length) control plant growth and development.
4. Trace the life cycle of a flowering plant from pollination to germination.
5. Place the major plant lineages on a phylogenetic tree and relate their features to life on land.
6. Connect plant biology to crop improvement, plant-based foods, and biotechnology, and read a simple plant science figure critically.

Prerequisites

General Biology (b)(1). Students take General Biology (b)(2) concurrently. No prior botany is assumed.

Textbooks and readings

- **Required (free, online):** OpenStax, *Biology 2e* (2018). Chapters 8 (Photosynthesis), 25 (Seedless Plants), 26 (Seed Plants), 30 (Plant Form and Physiology), 31 (Soil and Plant Nutrition), 32 (Plant Reproduction). https://openstax.org/details/books/biology-2e
- **Reference:** Evert RF, Eichhorn SE. *Raven Biology of Plants*, 8th ed. W.H. Freeman, 2013.
- **Reference:** Mauseth JD. *Botany: An Introduction to Plant Biology*, 7th ed. Jones & Bartlett, 2021.
- **Reference (advanced):** Taiz L, Møller IM, Murphy A, Zeiger E. *Plant Physiology and Development*, 7th ed. Oxford University Press, 2022.

Weekly reading assignments are posted on NTU COOL one week ahead.

Weekly schedule (16 weeks)

Periods 6–7 are lecture; period 8 is studio unless noted. Holiday dates are pending confirmation against the official NTU 115 academic calendar.

Wk	Date	Topic	Studio (period 8)	Reading
1	2/22	Why plants? Plant diversity at a glance. The plant cell: cell wall, plastids, vacuole, plasmodesmata	Course overview; team formation; "plant on your plate" icebreaker	OpenStax 4 (review), 30.1
2	3/1	**No class** (228 Peace Memorial Day, observed; to confirm)	—	—
3	3/8	Tissues and the plant body: meristems; dermal, ground, and vascular tissues; parenchyma, collenchyma, sclerenchyma, xylem, phloem cell types	Hand-cut sections of celery and leek; tissue identification worksheet	OpenStax 30.1–30.2
4	3/15	Roots, stems, and leaves: primary structure and function; leaf anatomy; stomata	Leaf epidermal peel and stomatal counts	OpenStax 30.2–30.4

Wk	Date	Topic	Studio (period 8)	Reading
5	3/22	Secondary growth and wood. Water relations: water potential, transpiration, cohesion–tension	Water potential problem set	OpenStax 30.5
6	3/29	Mineral nutrition and soils; nitrogen fixation and mycorrhizae; phloem translocation (pressure flow)	**Quiz 1** (Wk 1–5); nutrient deficiency gallery	OpenStax 31
7	4/5	**No class** (Tomb Sweeping Day)	—	—
8	4/12	Photosynthesis: light reactions, Calvin cycle, photorespiration, C4 and CAM; respiration and carbon allocation	Midterm review	OpenStax 8
9	4/19	**Midterm exam** (Wk 1–8; periods 6–7)	Exam debrief	—
10	4/26	Plant hormones: auxin, cytokinin, gibberellin, abscisic acid, ethylene, brassinosteroids, jasmonate, salicylic acid	Case studies: fruit ripening, seedless fruit, the Green Revolution	OpenStax 30.6
11	5/3	Sensing the environment: phytochrome, cryptochrome, phototropin; photoperiodism and flowering; circadian clock; tropisms	Design an experiment: how does a plant know it is spring?	OpenStax 30.6
12	5/10	Reproduction: flower structure, pollination, double fertilization, seed and fruit development, germination and dormancy	Flower dissection	OpenStax 32
13	5/17	Plant diversity and evolution: green algae to land plants; bryophytes, lycophytes, ferns, gymnosperms, angiosperms; alternation of generations	Build a plant phylogeny from traits	OpenStax 25–26
14	5/24	Plants and their environment: drought, salt, and temperature stress; pathogens, herbivores, and symbioses; plant immunity basics	**Quiz 2** (Wk 10–13)	Handout
15	5/31	Plants, food, and biotechnology: domestication, breeding, transgenic and genome-edited crops, plant-based foods	**Group poster session**	Handout

Wk	Date	Topic	Studio (period 8)	Reading
16	6/7	**Final exam** (Wk 10–15 plus integrative questions; periods 6–7)	Course wrap-up	—

Assessment

Component	Weight	Notes
Midterm exam	30%	Wk 9. Short answer and figure interpretation.
Final exam	30%	Wk 16. Emphasis on Wk 10–15 with integrative questions across the course.
Quizzes and studio worksheets	20%	Two quizzes (5% each) and 10 studio worksheets (1% each).
Group poster: "Plant on your plate"	10%	Teams of 3–4 pick a food or crop plant and present its biology from cell to field. Posters shown in Wk 15.
Participation	10%	Engagement in studio activities and discussion.

Course policies

- **Language:** All lectures, materials, and assessments are in English.
- **Materials:** Slides and readings go on NTU COOL the day before class.
- **Late work:** Studio worksheets are due at the end of the studio period. Other work loses 10% per day late, up to 3 days.
- **Generative AI:** You may use AI tools to study, outline, and polish English. State how you used them in any submitted work. AI tools are not allowed during quizzes and exams.
- **Academic integrity:** Follow NTU regulations. Cite sources in the group poster.
- **Mid-semester feedback:** Anonymous survey in Wk 8; the instructor responds in Wk 9.

設計理由與備課備忘（中文，自用）

定位

- 這是學程裡植物相關課程的入口，對象是大一下、以國際生為主、同時修普通生物學（乙）二。課程要做的是建立「整株植物」的圖像，不是植物生理的進階版。深度對標 OpenStax／

Raven，而不是 Taiz。

- 每週三節：兩節講課、一節 studio。studio 是讓大一學生動手與互動的機會，也是全英語課堂裡讓程度較弱的學生跟上的緩衝。第一年若負荷過重，studio 可退化為小考＋習題，不影響架構。

進度安排的考量

- 週一只有 14 次上課（3/1、4/5 停課，前者待確認）。期中排在第 9 週（4/19），期末在第 16 週（6/7），各佔兩節，第三節用來檢討或總結。
- 順序：細胞 → 組織 → 器官 → 運輸 → 營養 → 光合（期中）→ 荷爾蒙 → 環境感知 → 生殖 → 多樣性 → 逆境與互動 → 應用。多樣性放到第 13 週而不是開頭，理由是先讓學生熟悉被子植物的結構，再談演化比較容易理解「哪些構造是陸生化的適應」。
- 第 5、6、8、10、14 週可直接降階改寫 115-1 植物生理與逆境反應（BIOT7035）的投影片。

Studio 需要的材料與場地

- 第 3、4、12 週需要新鮮植物材料（芹菜、青蔥、葉片、花）、放大鏡或 Foldscope、指甲油與透明膠帶（氣孔印模）。請確認教室是否允許，若是固定座位的階梯教室，就改為示範影像＋學習單。
- 第 15 週海報：若修課人數超過 40 人，改為線上海報（NTU COOL 討論區）加 3 分鐘口頭簡介。

待確認

- 課號、學分（假設 3）、必修／選修、人數上限、教室型態。
- 3/1 是否補假（228 逢週日）。
- 學程是否要求大一課程配合特定評量規範（例如期中考週統一時間）。

Organelle Biology 胞器生物學

115-2 · Course code TBD · 2 credits (assumed) · Required/Elective TBD · Fri 6–7 (13:20–15:10) · Taught in English

Bachelor Program of Biotechnology and Food Nutrition (BFN) · Year 2 · Instructor: Yueh Cho

Course description

A eukaryotic cell is a city of compartments. This course takes each organelle in turn and asks four questions: how is it built, how do proteins and lipids get there, what does it do, and what happens when it fails? We cover the nucleus, endoplasmic reticulum, Golgi apparatus, endosomes and lysosomes or vacuoles, mitochondria, chloroplasts and other plastids, peroxisomes, and lipid droplets, and finish with how organelles talk to each other and how biotechnology exploits them. Examples come from yeast, animal, and plant cells, with links to nutrition, disease, and food science throughout.

Learning outcomes

By the end of the course, you will be able to:

1. Describe the structure, composition, and main functions of each major organelle.
2. Explain the logic of protein sorting: signal sequences, translocation machineries, and vesicle traffic.
3. Compare the endosymbiotic organelles (mitochondria and plastids) with the endomembrane system in origin, genome, and import mechanism.
4. Relate organelle quality control (ER stress, autophagy, mitophagy) to cell survival and disease.
5. Interpret fluorescence and electron micrographs, fractionation data, and simple organelle proteomics.
6. Give examples of organelle engineering in biotechnology and of organelle dysfunction in human nutrition and disease.

Prerequisites

General Biology (b)(1)(2). Concurrent or prior biochemistry is helpful but not required; the course reviews membrane structure in Wk 2.

Textbooks and readings

- **Required:** Alberts B, Hopkin K, Johnson A, Morgan D, Raff M, Roberts K, Walter P. *Essential Cell Biology*, 5th ed. W.W. Norton, 2019. Chapters 11 (Membrane Structure), 12 (Transport Across Cell Membranes), 14 (Energy Generation in Mitochondria and Chloroplasts), 15 (Intracellular Compartments and Protein Transport), 17 (Cytoskeleton).
- **Reference (advanced):** Alberts B et al. *Molecular Biology of the Cell*, 7th ed. W.W. Norton, 2022. Chapters 10, 12, 13, 14, 16.
- Selected review articles posted on NTU COOL for Wk 5, 9, 12, 14, and 15.

Weekly schedule (16 weeks)

Holiday dates are pending confirmation against the official NTU 115 academic calendar.

Wk	Date	Topic	Reading	Due
1	2/26	Why compartments? Endosymbiotic origin of mitochondria and chloroplasts; organelle census; methods: fluorescence and electron microscopy, subcellular fractionation, organelle proteomics	ECB 1, 15 (intro)	—
2	3/5	Membranes as organelle boundaries: lipid bilayer, membrane proteins, asymmetry, curvature; transporters and channels in brief	ECB 11, 12	—
3	3/12	The nucleus: nuclear envelope, nuclear pore complex, nucleocytoplasmic transport, nucleolus, chromatin domains, lamina	ECB 15	—
4	3/19	Protein sorting I: signal sequences and the sorting logic; co-translational translocation into the ER (SRP, Sec61); post-translational import into mitochondria (TOM/TIM), chloroplasts (TOC/TIC), and peroxisomes (PEX)	ECB 15	Problem set 1
5	3/26	The endoplasmic reticulum: folding and chaperones, N-glycosylation, ER quality control, ERAD, the unfolded protein response; the ER as a lipid factory	ECB 15; review	—

Wk	Date	Topic	Reading	Due
6	4/2	Golgi apparatus and vesicle traffic: COPII and COPI, cisternal maturation, glycan processing, sorting at the trans-Golgi network, clathrin, SNAREs and Rabs	ECB 15	—
7	4/9	Exocytosis, endocytosis, and endosomes: regulated secretion, receptor-mediated endocytosis, ESCRT and multivesicular bodies, recycling; plant-specific traffic	ECB 15	Problem set 2
8	4/16	**Midterm exam** (Wk 1–7)	—	—
9	4/23	Lysosomes and vacuoles: acidification, hydrolases, macroautophagy, microautophagy, chaperone-mediated autophagy, lysosomal storage diseases; plant vacuoles in storage, turgor, and pigmentation	ECB 15; review	—
10	4/30	**No class** (Labor Day observed; to confirm). If class is held: journal club on an organelle paper	—	—
11	5/7	Mitochondria I: architecture, genome and dual genetic control, electron transport chain, ATP synthase, metabolite carriers	ECB 14	Problem set 3
12	5/14	Mitochondria II: fission and fusion, mitophagy, mitochondrial unfolded protein response, apoptosis, reactive oxygen species; mitochondrial disease and nutrition	review	—
13	5/21	Chloroplasts and plastids: thylakoid architecture, plastid genome, plastid differentiation (chromoplasts, amyloplasts: color and starch in food), retrograde signaling	ECB 14	—
14	5/28	Peroxisomes and lipid droplets: biogenesis, β-oxidation, glyoxysomes, ROS metabolism; lipid droplet formation, oleosins and perilipins; membrane contact sites	review	Problem set 4
15	6/4	Organelle positioning and inheritance: cytoskeleton and motors, organelle division; organelle engineering in biotechnology (chloroplast transformation, organelle targeting, mitochondria-targeted therapeutics)	ECB 17; review	—
16	6/11	**Final exam** (Wk 9–15 plus integrative questions)	—	—

Assessment

Component	Weight	Notes
Midterm exam	30%	Wk 8. Short answer, pathway diagrams, micrograph interpretation.
Final exam	30%	Wk 16. Emphasis on Wk 9–15 with integrative questions.
Problem sets	25%	Four sets, due Wk 4, 7, 11, 14. Each mixes concept questions with a short data-interpretation task.
Participation	15%	In-class questions, polls, and the Wk 10 journal club if held.

Course policies

- **Language:** All lectures, materials, and assessments are in English.
- **Materials:** Slides and readings go on NTU COOL the day before class.
- **Late work:** Problem sets lose 10% per day late, up to 3 days.
- **Generative AI:** You may use AI tools to study and to polish English. State how you used them in any submitted work. AI tools are not allowed during exams.
- **Academic integrity:** Follow NTU regulations.
- **Mid-semester feedback:** Anonymous survey in Wk 8; the instructor responds in Wk 9.

設計理由與備課備忘（中文，自用）

定位

- 大二下，學生修過普生（乙）一、二與微生物學，可能正在修生化。這門課把「細胞」拆成胞器逐一講，是大一植物生物學（整株）與大三植物脂質（分子）之間的中間尺度，也是大三／四蛋白質恆定概論的前置課。
- 不限植物：BFN 的學生未必走植物路線，所以每個胞器都給酵母、動物、植物三種例子，並且每週掛一個營養或疾病的連結（粒線體與代謝、溶體儲積症、植體與食物色素／澱粉）。
- 教科書選 Essential Cell Biology 而不是 MBoC，原因是大二全英語課，ECB 的篇幅與難度較合適；MBoC 列為進階參考。

進度安排的考量

- 週五 6–7 節，15 次上課（4/30 待確認）。期中第 8 週、期末第 16 週，各佔整堂。

- 順序：膜 → 核 → 運送邏輯 → ER → Golgi → 胞吞／胞吐 → 溶體／液胞 → 粒線體 ×2 → 葉綠體 → 過氧化體與脂滴 → 骨架與工程。把 ER 放在第 5 週、剛好在期中前，是你最熟的主題，可當示範講得深一點；粒線體用兩週，因為營養相關內容最多。
- 第 5、9 週可改寫 115-1 蛋白質恆定概論（BFN5013）第 7、9、12 週的投影片，降低深度。第 13 週可借用植物生理的光合作用投影片。
- 第 14 週（過氧化體、脂滴、膜接觸點）是為下學年大三植物脂質鋪路，也直接對應你的第二個研究主題。

與 **Plant Lipids** 連排的實務

- 本課 15:10 結束，Plant Lipids 15:30 開始，中間 20 分鐘。建議兩門課用同一間教室；若不同教室，請確認距離。
- 第 8 週期中考若用整堂 100 分鐘，收卷後要立刻趕下一門，請安排助教收卷。

待確認

- 課號、學分（假設 2）、必修／選修、人數上限。
- 4/30 是否放假。
- 大二是否已有必修生化，決定第 2 週膜結構的深度。

Plant Lipids 植物脂質

115-2 · Course code TBD · 2 credits (assumed) · Required/Elective TBD · Fri 8–9 (15:30–17:20) · Taught in English

Bachelor Program of Biotechnology and Food Nutrition (BFN) · Year 3 · Instructor: Yueh Cho

Course description

Lipids make up every membrane in a plant, store the energy in every seed, waterproof every leaf, and carry signals that decide whether a plant grows, defends itself, or dies. This course follows plant lipids from fatty acid synthesis in the plastid to the oils in your kitchen. We cover the biosynthesis of membrane glycerolipids, sphingolipids, and sterols; storage lipids and oil bodies; surface lipids; lipid signaling in development, stress, and immunity; the methods of modern lipidomics; and the engineering of plant oils for food, nutrition, and industry.

The primary literature for the course is built around one research program: the work of Yuki Nakamura's Plant Lipid Research Team (RIKEN CSRS, formerly Academia Sinica), which over two decades traced how Arabidopsis makes, remodels, and uses its glycerolipids. Following one lab's papers lets you watch a research question evolve from the first phosphate-starvation phospholipase (2005) to lipid-regulated flowering (2014) and ER homeostasis (2024). Students read primary literature each week and present one research paper in the final weeks.

Learning outcomes

By the end of the course, you will be able to:

1. Draw the major plant lipid classes and explain where in the cell each is made.
2. Trace fatty acid synthesis, desaturation, and glycerolipid assembly through the prokaryotic and eukaryotic pathways, and explain how lipids move between the plastid and the ER.
3. Explain how membrane lipid composition changes with temperature, phosphate supply, and stress, and why.
4. Describe triacylglycerol synthesis, oil body formation, and lipid mobilization in seeds.
5. Outline the main lipid signaling pathways (phosphatidic acid, phosphoinositides, oxylipins, sphingolipids) and their roles in stress and immunity.

6. Read and critique a lipidomics dataset, and evaluate a plant oil engineering strategy for food or industrial use.

Prerequisites

General Biology and Biochemistry. Organelle Biology or Plant Physiology is helpful; Wk 1–2 review the membrane and organelle background needed.

Textbooks and readings

- **Required chapters:** Buchanan BB, Gruissem W, Jones RL (eds). *Biochemistry & Molecular Biology of Plants*, 2nd ed. Wiley-Blackwell, 2015. Chapter 8 (Lipids).
- **Required review (open access):** Li-Beisson Y, Shorrosh B, Beisson F, et al. Acyl-lipid metabolism. *The Arabidopsis Book* 11: e0161 (2013).
- **Reference:** Heldt H-W, Piechulla B. *Plant Biochemistry*, 5th ed. Academic Press, 2021. Lipid chapter.
- **Primary literature:** the week-by-week Nakamura lab reading list in `plant-lipids-nakamura-reading-list.md` (required, supplementary, and presentation papers marked per week).
- **Selected reviews (posted on NTU COOL):**
    - Ohlrogge J, Browse J. Lipid biosynthesis. *Plant Cell* 7: 957–970 (1995).
    - Hölzl G, Dörmann P. Chloroplast lipids and their biosynthesis. *Annual Review of Plant Biology* 70: 51–81 (2019).
    - Nakamura Y. Plant phospholipid diversity: emerging functions in metabolism and protein–organelle interactions. *Trends in Plant Science* 22: 1027–1040 (2017).
    - Nakamura Y. Headgroup biosynthesis of phosphatidylcholine and phosphatidylethanolamine in seed plants. *Progress in Lipid Research* 82: 101091 (2021).
    - Nakamura Y. Phosphate starvation and membrane lipid remodeling in seed plants. *Progress in Lipid Research* 52: 43–50 (2013).
    - Nakamura Y, Ngo AH. Non-specific phospholipase C (NPC): an emerging class of phospholipase C in plant growth and development. *Journal of Plant Research* 133 (2020).
    - Bates PD, Stymne S, Ohlrogge J. Biochemical pathways in seed oil synthesis. *Current Opinion in Plant Biology* 16: 358–364 (2013).

- Kanehara K, Yu C-Y, Cho Y. A lipid viewpoint on the plant endoplasmic reticulum stress response. *Journal of Experimental Botany* 73: 2835–2847 (2022).
- Additional reviews on sphingolipids, sterols, and cutin and suberin will be added to NTU COOL before the relevant week.

Weekly schedule (16 weeks)

Holiday dates are pending confirmation against the official NTU 115 academic calendar.

Wk	Date	Topic	Reading	Due
1	2/26	Introduction: lipid classes and nomenclature; where lipids live in the plant cell; why plant lipids matter for membranes, energy, signaling, food, and industry	BGJ 8 (intro); Li-Beisson §1; Nakamura 2017 TIPS	—
2	3/5	Fatty acid synthesis in the plastid: acetyl-CoA carboxylase, type II fatty acid synthase, acyl carrier protein, stearoyl-ACP desaturase, fatty acid export	BGJ 8; Li-Beisson §2	—
3	3/12	Glycerolipid assembly: the Kennedy pathway; prokaryotic versus eukaryotic pathways; phosphatidic acid phosphatases and acyltransferases; lipid trafficking between the ER and plastid (TGD complex)	Ohlrogge & Browse 1995; Nakamura et al. 2009 PNAS; Nguyen & Nakamura 2023 Plant Cell	Reading response 1
4	3/19	Chloroplast membrane lipids: MGDG, DGDG, SQDG, PG; roles in thylakoid structure and photosynthesis; membrane remodeling under phosphate starvation (NPC4, NPC5, lipins)	Hölzl & Dörmann 2019; Nakamura 2013 PLR; Nakamura et al. 2005 JBC	—
5	3/26	Phospholipids of the extraplastidial membranes: PC, PE, PI, PS, PA; CDP-DAG and CDP-choline routes; choline/ethanolamine kinases, phospho-base methyltransferases, and PLMT; phosphatidylcholine as a metabolic hub	Nakamura 2021 PLR; Lin et al. 2015 Plant Cell	Problem set 1

Wk	Date	Topic	Reading	Due
6	4/2	Fatty acid desaturation and modification: FAD2 and FAD3, very-long-chain fatty acid elongation, unusual fatty acids; membrane fluidity, temperature acclimation, and ER stress tolerance	Li-Beisson §4; Nguyen et al. 2019 Plant J	—
7	4/9	Sphingolipids and sterols: structures, biosynthesis, membrane nanodomains, roles in development and programmed cell death	review (TBA)	Reading response 2
8	4/16	**Midterm exam** (Wk 1–7)	—	—
9	4/23	Storage lipids: triacylglycerol synthesis (DGAT, PDAT, phospholipid-derived routes), oil bodies and oleosins, seed oil accumulation, lipid droplets in pollen tubes; lipid mobilization: β-oxidation and the glyoxylate cycle; algal oil	Bates et al. 2013; Bose et al. 2021 Plant J	—
10	4/30	**No class** (Labor Day observed; to confirm). If class is held: paper selection workshop for final presentations	—	—
11	5/7	Surface lipids: cutin, suberin, and cuticular waxes; biosynthesis, ABCG transporters, and barrier functions in water loss and defense	review (TBA)	Problem set 2
12	5/14	Lipid signaling I: phospholipases D, C (PI-PLC and non-specific PLC), and A; phosphatidic acid and phosphoinositides in stress and development	Nakamura & Ngo 2020 JPR; Kanehara et al. 2015 PLoS Genet	—
13	5/21	Lipid signaling II: oxylipins and jasmonate; lipids in plant immunity; lipids in flowering (florigen binds phosphatidylcholine); lipids as regulators of ER homeostasis and the unfolded protein response	Nakamura et al. 2014 Nat Commun; Kanehara, Yu & Cho 2022	Reading response 3

Wk	Date	Topic	Reading	Due
14	5/28	Lipid analysis: extraction, thin-layer chromatography, GC-FID and GC-MS fatty acid profiling, LC-MS lipidomics, isotope labeling, lipid imaging; data-reading workshop on published lipidomic tables	handout; Nakamura et al. 2014 New Phytol (data)	—
15	6/4	Plant oils for food, nutrition, and industry: high-oleic and omega-3 oilseed engineering, phospholipid routes to seed oil, biofuels and algal oil, fatty acid composition and human health; **paper presentations I**	handout; Tan & Nakamura 2022 JXB	Presentation slides
16	6/11	**Paper presentations II** and course wrap-up	—	Presentation slides

Assessment

Component	Weight	Notes
Midterm exam	30%	Wk 8. Pathway diagrams, short answer, and figure interpretation.
Final paper presentation	30%	Pairs present a primary research paper from the Nakamura lab candidate list (15 min talk plus 5 min questions) in Wk 15–16. Graded on scientific accuracy, clarity, and handling of questions.
Problem sets and reading responses	25%	Two problem sets (7.5% each) and three one-page reading responses (3.3% each).
Participation	15%	In-class discussion and questions during peer presentations.

Course policies

- **Language:** All lectures, materials, and assessments are in English.
- **Materials:** Slides and readings go on NTU COOL the day before class.
- **Late work:** Problem sets and reading responses lose 10% per day late, up to 3 days.
- **Generative AI:** You may use AI tools to study, to summarize papers for your own notes, and to polish English. State how you used them in any submitted work. AI tools are not allowed during the exam.

- **Academic integrity:** Follow NTU regulations. Presentations must cite the paper and all figure sources.
- **Mid-semester feedback:** Anonymous survey in Wk 8; the instructor responds in Wk 9.

設計理由與備課備忘（中文，自用）

定位

- 大三下的專門課，是三門課裡最接近你專業的一門（Nakamura 實驗室背景、JXB 2022 lipid viewpoint）。可以直接用你的研究當案例，也最適合先做完投影片，建立三門課的製作節奏。
- 對象是 BFN 大三，食品營養是學程主軸之一，因此第 15 週把油脂工程與人類營養放在一起講，並且全程用「食用油的脂肪酸組成」當反覆出現的例子。
- 評量以文獻報告取代期末筆試：大三學生應該開始讀原始文獻，這也是 115-1 蛋白質恆定概論用分組報告當期末的同一邏輯。

進度安排的考量

- 週五 8–9 節，15 次上課（4/30 待確認）。合成（第 2–7 週）→ 期中 → 儲存、表面、訊號（第 9–13 週）→ 方法與應用（第 14–15 週）→ 報告。
- 第 1–2 週務必補膜與胞器的基礎：115-2 修這門課的大三生沒修過胞器生物學。從 116 學年起這段可縮短。
- 第 3 週「原核／真核途徑」與第 5 週「PC 為代謝樞紐」是最容易讓學生卡住的兩週，各配一份問題集或閱讀回應。
- 第 13 週直接講你的研究：脂質如何調控 ER 壓力反應（Kanehara, Yu & Cho 2022；PLC2 與 ER 壓力的 PLoS Genetics 2015）。這週也是招募專題生的機會。
- 第 14 週的資料判讀工作坊可用實驗室的 lipidomics 資料（去識別化後）當教材。

以 **Nakamura** 實驗室的發表為主軸

- 依你的指示，原始文獻改以 Yuki Nakamura 的著作整合，逐週對應放在 `plant-lipids-nakamura-reading-list.md`，每週標 R（必讀）、S（補充）、P（報告候選），期末報告候選 18 篇也在該檔。
- 覆蓋狀況：第 3、4、5、9、12、13 週有大量對應，第 6、14、15 週各有 1–2 篇；第 2（脂肪酸合成）、7（鞘脂與固醇）、11（表面脂質）週沒有他的論文，維持教科書與回顧文獻。
- 這個環境無法連 PubMed、Europe PMC、Scholar 與 RIKEN 網站，清單只收錄我能確認或有把握的約 50 篇（他的著作約 90 篇）。每條有 [V]（本次驗證過）或 [M]（記憶，待核對）標記；上線前請用 Scholar 頁面補齊並核對 [M] 的卷期頁碼。

- 第 13 週同時用 Nakamura 2014 Nat Commun（FT 結合 PC）與你們 2022 JXB 的 lipid viewpoint，可以把「從 Nakamura 實驗室到你實驗室」的研究脈絡講成一條線，也是招募專題生最自然的時機。

與 **Organelle Biology** 連排的實務

- 前一門課 15:10 結束，本課 15:30 開始。建議本課前 10 分鐘固定為學生自行閱讀當週指定文獻或作答一題暖身題，給自己換場與喘息的時間。

待確認

- 課號、學分（假設 2）、必修／選修、人數上限。
- 4/30 是否放假。
- 大三是否已修必修生化（決定第 2 週脂肪酸合成講多細）。
- 修課人數：若超過 16 人，報告改為三人一組或縮短為 10 分鐘，否則第 15–16 週排不完。

Plant Lipids: Reading list built on the Nakamura lab publications

> 給自己的說明（2026-10-24）：這份清單把 Yuki Nakamura（RIKEN CSRS 植物脂質研究團隊，前中研院植微所）的發表依 Plant Lipids 的週次整合。
> 這個環境無法連 PubMed、Europe PMC、Google Scholar 與 RIKEN 網站，所以清單只涵蓋我能確認或有把握的約 50 篇，不是他全部約 90 篇著作。
> 每條標記：**[V]** 本次由搜尋結果或你的 CV 確認過書目；**[M]** 出自我的記憶，書目細節請在上線前核對。
> 補齊其餘著作請用：Google Scholar 個人頁、RIKEN Plant Lipid Research Team、PubMed 查詢 `Nakamura Y[au] AND (Academia Sinica[ad] OR RIKEN[ad]) AND lipid`。
> 尚未對應到 Nakamura 論文的週次（第 2、7、11 週）維持原本的教科書與回顧文獻。

Legend for students: **R** = required reading for the week; **S** = supplementary; **P** = candidate paper for the final presentation.

Wk 1 · Introduction: lipid diversity and why it matters

- **R** [V] Nakamura Y. Plant phospholipid diversity: emerging functions in metabolism and protein–organelle interactions. *Trends Plant Sci* 22: 1027–1040 (2017).
- **S** [V] Nakamura Y. Membrane lipid oscillation: an emerging system of molecular dynamics in the plant membrane. *Plant Cell Physiol* 59: 441–447 (2018).
- **S** [M] Nakamura Y, Li-Beisson Y (eds). *Lipids in Plant and Algae Development*. Subcellular Biochemistry vol. 86. Springer (2016).
- **S** [V] Kilaru A, Allen DK, Yandeau-Nelson MD, Nakamura Y. Lipid research toward advancing plant performance and environmental sustainability. *J Exp Bot* 77: 3263– (2026). Special-issue editorial; confirm author order.

Wk 3 · Glycerolipid assembly, phosphatidic acid phosphatases, and ER–plastid cooperation

- **R** [V] Nakamura Y, Koizumi R, Shui G, Shimojima M, Wenk MR, Ito T, Ohta H. Arabidopsis lipins mediate eukaryotic pathway of lipid metabolism and cope critically with phosphate

starvation. *Proc Natl Acad Sci USA* 106: 20978–20983 (2009). **P**

- **R** [V] Nguyen VC, Nakamura Y. Distinctly localized lipid phosphate phosphatases mediate endoplasmic reticulum glycerolipid metabolism in Arabidopsis. *Plant Cell* 35: 1548–1571 (2023). **P**
- **S** [V] Nakamura Y, Tsuchiya M, Ohta H. Plastidic phosphatidic acid phosphatases identified in a distinct subfamily of lipid phosphate phosphatases with prokaryotic origin. *J Biol Chem* 282: 29013–29021 (2007).
- **S** [M] Angkawijaya AE, Nguyen VC, Nakamura Y. Enhanced root growth in phosphate-starved Arabidopsis by stimulating de novo phospholipid biosynthesis through the overexpression of LYSOPHOSPHATIDIC ACID ACYLTRANSFERASE 2 (LPAT2). *Plant Cell Environ* 40: 1807–1818 (2017). **P**
- **S** [M] Angkawijaya AE, Nguyen VC, Nakamura Y. LYSOPHOSPHATIDIC ACID ACYLTRANSFERASES 4 and 5 are involved in glycerolipid metabolism and nitrogen starvation response in Arabidopsis. *New Phytol* 224: 336–351 (2019).
- **S** [V] Barroga NAM, Nguyen VC, Nakamura Y. The role of lysophosphatidic acid acyltransferase 1 in reproductive growth of Arabidopsis thaliana. *J Exp Bot* 75: 7190–7201 (2024). **P**
- **S** [V] Angkawijaya AE, Nguyen VC, Gunawan F, Nakamura Y. A pair of Arabidopsis diacylglycerol kinases essential for gametogenesis and endoplasmic reticulum phospholipid metabolism in leaves and flowers. *Plant Cell* 32: 2602–2620 (2020). **P**
- **S** [M] Nakamura Y, Ohta H. The diacylglycerol forming pathways differ among floral organs of Petunia hybrida. *FEBS Lett* 581: 5475–5479 (2007).

Wk 4 · Chloroplast lipids and membrane remodeling under phosphate starvation

- **R** [V] Nakamura Y. Phosphate starvation and membrane lipid remodeling in seed plants. *Prog Lipid Res* 52: 43–50 (2013).
- **R** [V] Nakamura Y, Awai K, Masuda T, Yoshioka Y, Takamiya K, Ohta H. A novel phosphatidylcholine-hydrolyzing phospholipase C induced by phosphate starvation in Arabidopsis. *J Biol Chem* 280: 7469–7476 (2005). **P**
- **S** [V] Gaude N, Nakamura Y, Scheible W-R, Ohta H, Dörmann P. Phospholipase C5 (NPC5) is involved in galactolipid accumulation during phosphate limitation in leaves of Arabidopsis. *Plant J* 56: 28–39 (2008).
- **S** [M] Angkawijaya AE, Nakamura Y. Arabidopsis PECP1 and PS2 are phosphate starvation-inducible phosphoethanolamine/phosphocholine phosphatases. *Biochem*

Biophys Res Commun 494: 397–401 (2017).

- **S** [V] Lin Y-C, Kobayashi K, Hung C-H, Wada H, Nakamura Y. Arabidopsis PHOSPHATIDYLGLYCEROPHOSPHATE PHOSPHATASE1 involved in phosphatidylglycerol biosynthesis and photosynthetic function. *Plant J* 88: 1022–1037 (2016). **P**
- **S** [V] Lin Y-C, Kobayashi K, Wada H, Nakamura Y. Phosphatidylglycerophosphate phosphatase is required for root growth in Arabidopsis. *Biochim Biophys Acta Mol Cell Biol Lipids* (2018). Volume and pages to confirm.
- **S** [M] Kobayashi K, Nakamura Y, Ohta H. Type A and type B monogalactosyldiacylglycerol synthases are spatially and functionally separated in the plastids of higher plants. *Plant Physiol Biochem* 47: 518–525 (2009).
- **S** [V] Nakamura Y, Kobayashi K, Ohta H. Activation of galactolipid biosynthesis in development of pistils and pollen tubes. *Plant Physiol Biochem* 47: 535–539 (2009).
- **S** [V] Nakamura Y. Galactolipid biosynthesis in flowers. *Botanical Studies* 54: 29 (2013).
- **S** [M] Nakamura Y, Arimitsu H, Yukawa Y, Ohta H. Digalactosyldiacylglycerol is a major glycolipid in floral organs of Petunia hybrida. *Lipids* 38: 1107–1112 (2003).

Wk 5 · Phospholipid headgroups: phosphatidylcholine and phosphatidylethanolamine biosynthesis

- **R** [V] Nakamura Y. Headgroup biosynthesis of phosphatidylcholine and phosphatidylethanolamine in seed plants. *Prog Lipid Res* 82: 101091 (2021).
- **R** [V] Lin Y-C, Liu Y-C, Nakamura Y. The choline/ethanolamine kinase family in Arabidopsis: essential role of CEK4 in phospholipid biosynthesis and embryo development. *Plant Cell* 27: 1497–1511 (2015). **P**
- **S** [M] Lin Y-C, Kanehara K, Nakamura Y. Arabidopsis CHOLINE/ETHANOLAMINE KINASE 1 (CEK1) is a primary choline kinase localized at the endoplasmic reticulum and involved in ER stress tolerance. *New Phytol* 223: 1904–1917 (2019). **P**
- **S** [M] Liu Y-C, Lin Y-C, Kanehara K, Nakamura Y. A methyltransferase trio essential for phosphatidylcholine biosynthesis and growth. *Plant Physiol* 179: 433–445 (2019). **P**
- **S** [V] Yunus IS, Liu Y-C, Nakamura Y. The importance of SERINE DECARBOXYLASE1 (SDC1) and ethanolamine biosynthesis during embryogenesis of Arabidopsis thaliana. *Plant J* 88: 559–569 (2016).
- **S** [V] Liu Y-C, Gunawan F, Yunus IS, Nakamura Y. Arabidopsis serine decarboxylase 1 (SDC1) in phospholipid and amino acid metabolism. *Front Plant Sci* 9: 972 (2018).
- **S** [V] Tan Y-R, Nakamura Y. The importance of Arabidopsis PHOSPHOLIPID N-METHYLTRANSFERASE (PLMT) in glycerolipid metabolism and plant growth. *J Exp Bot*

(2022). doi:10.1093/jxb/erac049. **P**

- **S** [V] Genetic defects in the CDP-choline pathway for phosphatidylcholine biosynthesis cannot be transmitted to offspring via male gametophytes owing to interruption of autophagy-like processes required for pollen germination in Arabidopsis thaliana (2026). PMC12992434. Authorship and journal to confirm before listing to students.

Wk 6 · Desaturation, membrane fluidity, and the ER

- **S** [M] Nguyen VC, Nakamura Y, Kanehara K. Membrane lipid polyunsaturation mediated by FATTY ACID DESATURASE 2 (FAD2) is involved in endoplasmic reticulum stress tolerance in Arabidopsis thaliana. *Plant J* 99: 478–493 (2019). **P**

Wk 9 · Storage lipids: triacylglycerol, lipid droplets, and algal oil

- **R** [V] Bose D, Ngo AH, Nguyen VC, Nakamura Y. Non-specific phospholipases C2 and C6 redundantly function in pollen tube growth via triacylglycerol production in Arabidopsis. *Plant J* 106: 409–418 (2021). **P**
- **S** [V] Tan Y-R, Nakamura Y. (2022) *J Exp Bot*, as in Wk 5: PLMT overexpression raises seed triacylglycerol.
- **S** [V] Hung C-H, Ho M-Y, Kanehara K, Nakamura Y. Functional study of diacylglycerol acyltransferase type 2 family in Chlamydomonas reinhardtii. *FEBS Lett* 587: 2364–2370 (2013).
- **S** [V] Hung C-H, Kanehara K, Nakamura Y. Isolation and characterization of a mutant defective in triacylglycerol accumulation in nitrogen-starved Chlamydomonas reinhardtii. *Biochim Biophys Acta* (2016). Title wording, volume, and pages to confirm.
- **S** [M] Hung C-H, Kanehara K, Nakamura Y. In vivo reconstitution of algal triacylglycerol production in Saccharomyces cerevisiae. *Front Microbiol* 7: 70 (2016).

Wk 12 · Lipid signaling I: phospholipases, phosphatidic acid, and phosphoinositides

- **R** [V] Nakamura Y, Ngo AH. Non-specific phospholipase C (NPC): an emerging class of phospholipase C in plant growth and development. *J Plant Res* 133 (2020). doi:10.1007/s10265-020-01199-8.
- **R** [V] Kanehara K, Yu C-Y, Cho Y, Cheong W-F, Torta F, Shui G, Wenk MR, Nakamura Y. Arabidopsis AtPLC2 is a primary phosphoinositide-specific phospholipase C in

phosphoinositide metabolism and the endoplasmic reticulum stress response. *PLoS Genet* 11: e1005511 (2015). **P**

- **S** [V] Ngo AH, Lin Y-C, Liu Y-C, Gutbrod K, Peisker H, Dörmann P, Nakamura Y. A pair of nonspecific phospholipases C, NPC2 and NPC6, are involved in gametophyte development and glycerolipid metabolism in Arabidopsis. *New Phytol* 219: 163–175 (2018). **P**
- **S** [V] Ngo AH, Kanehara K, Nakamura Y. Non-specific phospholipases C, NPC2 and NPC6, are required for root growth in Arabidopsis. *Plant J* 100: 825–835 (2019).
- **S** [M] Nakamura Y. NPC: nonspecific phospholipase Cs in plant growth and development. In: Wang X (ed) *Phospholipases in Plant Signaling*. Springer (2014). The 2020 review cites a 2014 Nakamura review; confirm this is it.
- **S** [M] Nakamura Y, Ohta H. Phosphatidic acid phosphatases in seed plants. In: Munnik T (ed) *Lipid Signaling in Plants*. Springer (2010). Confirm.

Wk 13 · Lipid signaling II: lipids in flowering, development, and ER homeostasis

- **R** [V] Nakamura Y, Andrés F, Kanehara K, Liu Y-C, Dörmann P, Coupland G. Arabidopsis florigen FT binds to diurnally oscillating phospholipids that accelerate flowering. *Nat Commun* 5: 3553 (2014). **P**
- **R** [V] Kanehara K, Yu C-Y, Cho Y. A lipid viewpoint on the plant endoplasmic reticulum stress response. *J Exp Bot* 73: 2835–2847 (2022). Instructor's own review; not a Nakamura paper.
- **S** [V] Nakamura Y, Teo NZ, Shui G, Chua CH, Cheong WF, Parameswaran S, Koizumi R, Ohta H, Wenk MR, Ito T. Transcriptomic and lipidomic profiles of glycerolipids during Arabidopsis flower development. *New Phytol* 203: 310–322 (2014).
- **S** [V] Nakamura Y. Function of polar glycerolipids in flower development in Arabidopsis thaliana. *Prog Lipid Res* 60: 17–29 (2015).
- **S** [V] Kanehara K, Cho Y, Lin Y-C, Chen C-E, Yu C-Y, Nakamura Y. Arabidopsis DOK1 encodes a functional dolichol kinase involved in reproduction. *Plant J* 81: 292–303 (2015).
- **S** [V] Cho Y, Yu C-Y, Nakamura Y, Kanehara K. Arabidopsis dolichol kinase AtDOK1 is involved in flowering time control. *J Exp Bot* 68: 3243–3252 (2017). **P**
- **S** [V] Ngo AH, Angkawijaya AE, Nakamura Y, Kanehara K. Non-specific phospholipase C3 is involved in endoplasmic reticulum stress tolerance in Arabidopsis. *J Exp Bot* 75: 6489–6499 (2024). **P**

- **S** [V] Yu C-Y, Nakamura Y. SMALLER TRICHOMES WITH VARIABLE BRANCHES (SVB) and its homolog SVBL act downstream of transcription factor NAC089 and function redundantly in Arabidopsis unfolded protein response. *J Exp Bot* 74: 5870–5880 (2023). **P**
- **S** [V] Ngo AH, Wu Y-C, Nakamura Y. Bidirectional movement of tunicamycin in Arabidopsis thaliana. *New Phytol* 241: 10–16 (2024).
- **S** [V] Yu C-Y, Cho Y, Sharma O, Kanehara K. What's unique? The unfolded protein response in plants. *J Exp Bot* 73: 1268–1276 (2022). Background on the UPR; not a Nakamura paper.

Wk 14 · Lipid analysis and data-reading workshop

- Dataset sources for the workshop: the lipidomic tables in Nakamura et al. *New Phytol* 203 (2014) and Nguyen & Nakamura *Plant Cell* 35 (2023); the PNAS 2009 lipin paper for thin-layer chromatography and fatty acid profiling by GC.

Wk 15 · Plant oils for food, nutrition, and industry

- **R** [V] Tan Y-R, Nakamura Y. (2022) *J Exp Bot*: a metabolic route from phosphatidylethanolamine to seed oil; overexpression raised seed triacylglycerol by about 20% per dry weight (figure from the RIKEN press release; confirm against the paper).
- **S** [M] Angkawijaya et al. (2017) *Plant Cell Environ*: engineering phospholipid synthesis to sustain root growth under low phosphate.
- **S** [V] Hung et al. (2013, 2016): Chlamydomonas triacylglycerol as an algal oil platform.

Candidate papers for the final presentation (one per pair; first come, first served on NTU COOL)

1. Nakamura et al. 2009 PNAS, lipins and the eukaryotic pathway
2. Nguyen & Nakamura 2023 Plant Cell, three classes of phosphatidic acid phosphatase
3. Angkawijaya et al. 2020 Plant Cell, diacylglycerol kinases and gametogenesis
4. Barroga et al. 2024 J Exp Bot, LPAT1 in reproductive growth
5. Nakamura et al. 2005 J Biol Chem, NPC4 and phosphate starvation
6. Lin et al. 2016 Plant J, PGPP1 and photosynthesis

7. Lin et al. 2015 Plant Cell, CEK4 and embryo development
8. Lin et al. 2019 New Phytol, CEK1 and ER stress tolerance
9. Liu et al. 2019 Plant Physiol, the methyltransferase trio
10. Tan & Nakamura 2022 J Exp Bot, PLMT and seed oil
11. Nguyen et al. 2019 Plant J, FAD2 and ER stress tolerance
12. Bose et al. 2021 Plant J, NPC2/NPC6 and pollen tube triacylglycerol
13. Kanehara et al. 2015 PLoS Genet, PLC2 and the ER stress response
14. Ngo et al. 2018 New Phytol, NPC2/NPC6 and gametophyte development
15. Nakamura et al. 2014 Nat Commun, florigen binds phosphatidylcholine
16. Cho et al. 2017 J Exp Bot, dolichol kinase and flowering time
17. Ngo et al. 2024 J Exp Bot, NPC3 and ER stress tolerance
18. Yu & Nakamura 2023 J Exp Bot, SVB/SVBL in the unfolded protein response